

Branch-and-Price Redistricting as an Audit-Ready Column-Generation Contract

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Draft — May 2026

Abstract

This paper presents the BISECT branch-and-price path as a staged exact optimization implementation. It describes pricing, master-problem, bound, exact fixture, and RPLAN/RCTX audit-package contracts without claiming mature large-instance branch-price performance. The paper treats disconnected columns, partial pools, and formulation-only outputs as reportable statuses rather than silent solver failures.

1 Introduction

U.17 documents `bisect-column` and `bisect-cli::exact_cmd`, exposed as `bisect exact --method branch-and-price`. The paper is an exact-optimization implementation paper about column-generation contracts, not a claim that the large-instance pricing strategy is complete.

The purpose is to make branch-and-price inspectable before production-scale solver integration. U.16 works with assignment variables and connectivity cuts; U.17 works with whole district columns and a set-partitioning master problem. U.20 then gives both families the same fixed-point destination: selected plans become RPLAN/RCTX artifacts, while solver-specific evidence stays in the sidecar reports and method lineage.

2 Implementation Contract

The branch-and-price surface has three contracts: a pricing contract that generates connected balanced columns or explicit failure statuses, a master contract that reports objective and bound state, and a package contract that turns solved fixture outputs into final-plan artifacts.

The key reproducibility requirement is that a reader can tell which loop stage produced the artifact. A pricing artifact is not a solved partition; a master report is not necessarily a branch-and-price proof; and an exact fixture package is scoped to the fixture model, objective, and column universe recorded in the run context.

In the atlas view, the candidates are whole district columns rather than individual unit moves. A low-cost column can still be rejected by the master if it overlaps selected columns

or leaves another unit uncovered. The selected solution must therefore name column ids and coverage status, not only the final assignment colors.

On the current path4 fixture, the graph has units $0 - 1 - 2 - 3$, equal populations, $k = 2$, and a one-percent tolerance. Pricing enumerates connected balanced subsets. The eligible district columns are $\{0, 1\}$, $\{1, 2\}$, and $\{2, 3\}$. The middle column is individually valid, but it cannot appear with either endpoint column without double-covering unit 1 or unit 2. The master therefore selects the endpoint columns, records selected column ids $[0, 2]$, emits assignment $[0, 0, 1, 1]$, and reports objective 1. This is the small example that separates column validity from exact cover validity.

Contract	Must report	Evidence level
Pricing	column validity, balance, reduced-cost status	L0 fixture
Master	selected columns, objective, bound status	L0/L1 exact package
Package	final assignments, manifest, certificate	L1 audit path

Table 1: Branch-and-price staged implementation contracts.

Report field	Current value family	Interpretation
Report schema	branch-price report v1	Stable JSON envelope for U.17 evidence
Status	formulation-only, exact-fixture-optimal, infeasible	Allowed claim vocabulary
Pricing counts	rounds and generated columns	Size of generated column pool
Bounds	lower bound, upper bound, gap	Master evidence when solved
Formulation size	constraints, columns, incidence nonzeros	Rebuildability and scale context
Solution	selected column ids, assignment, objective	Bridge from columns to final RPLAN
Parameter hash	SHA-256 digest	Tamper check for report-defining parameters

Table 2: Concrete branch-price report fields emitted by `bisect-column`.

The status table intentionally uses a richer explanatory vocabulary than the current enum. The implementation enum has `formulation-only`, `exact-fixture-optimal`, and `infeasible`. The paper-level terms “pricing-valid,” “master-bounded,” and “fixture-solved” are reader-facing states that explain which part of the lifecycle produced the evidence.

Status	Meaning	Allowed claim
Pricing-valid	connected balanced columns generated	pricing contract works on fixture
Pricing-failed	no valid column or explicit fall-back	failure is reproducible
Master-bounded	objective and bound reported	scoped bound under column pool
Fixture-solved	exact fixture package emitted	exact fixture result
Formulation-only	model/report built without solve	staged implementation only

Table 3: Status vocabulary for branch-and-price outputs.

3 Audit Artifacts

Solved outputs should converge to the RPLAN/RCTX fixed point with certificate and manifest sidecars. Column-generation-specific evidence remains in the run context and solver report; the final RPLAN carries the assignment, not the entire proof history.

The audit package should identify the column pool, selected columns, master objective, fixture or instance id, and certificate profile. This keeps the generic RPLAN assignment separate from the specialized column-generation evidence needed to interpret exactness.

The public golden package can be checked either through the RPLAN certificate verifier or through the BISECT manifest verifier.

The fixed point is deliberately two-layered. The RPLAN/RCTX pair fixes the assignment and input graph; the branch-price report fixes the column-generation evidence that led to the assignment. The algorithm lineage includes producer `bisect-column`, method `branch-and-price`, report schema, status, pricing rounds, generated column count, bounds, gap, and parameter hash. Editing the selected columns or changing a formulation-only run into a solved exact claim is therefore not an innocent prose change; it changes the sidecar evidence that the package is expected to preserve.

4 Evaluation Plan

The near-term evidence is L0 pricing/master tests, an L1 exact package test, and a public U.17 golden package in the package corpus. Comparisons with branch-and-cut, formulation-only reports, and heuristic construction are future work until command transcripts and larger fixture outputs are attached.

The CLI shape for the exact fixture path is:

```
bisect exact --method branch-and-price \
  --context fixture.rctx --out-dir exact \
  --districts 2 --tolerance 1.0
```

Passing `--formulation-only` changes the claim: the output can document the generated master but must not be described as a solved exact plan.

Claim			Current evidence	Publication gap
Pricing	emits	valid	L0 pricing fixture tests	column-pool transcript
columns				
Master	reports	objec-	L0/L1 master package tests	solver version and output
tive/bound				log
Solved	outputs	audit	L1 exact package test and pub-	larger solver-produced in-
through RPLAN			lic golden package	stance
Large-instance	branch-		future evidence placeholder	benchmark protocol
price scales				

Table 4: Evidence ladder for U.17 claims.

Command or package	Evidence produced
<code>cargo test -p bisect-column</code>	Checks pricing, formulation-only reports, exact path4 master enumeration, bounds, selected columns, and lineage hashes.
<code>bisect exact</code> fixture run	Writes branch-price report, algorithm-lineage JSON, exact RPLAN/RCTX, audit certificate, and manifest when a tiny exact solution exists.
Public golden package	Preserves the U.17 method transcript, plan, context, manifest, and audit certificate in the package corpus.

Table 5: Current U.17 evidence path.

5 Limitations

The current branch-and-price story is intentionally fixture-scale. Branching strategy, large-instance pricing, and broad runtime claims remain staged. The paper should not describe a formulation-only report or partial column pool as a solved exact instance.

The paper also does not claim that the generated column universe is complete unless the solver report and fixture contract say so. Pricing failures, duplicate columns, and disconnected candidates must remain visible because they are part of the implementation evidence.

The path4 fixture is useful because the exact-cover decision is inspectable by hand, but that same smallness limits the claim. A future performance paper needs larger formulation-only examples, solver versions, hardware, full logs, pricing-round transcripts, and archived output directories before comparing branch-and-price with branch-and-cut or construction heuristics.

6 Conclusion

U.17 establishes the column-generation audit contract: pricing and master artifacts may be specialized, but selected final plans should still reach the same RPLAN/RCTX verification path as other U-series algorithms.

This lets the implementation mature in public stages. The paper can claim a fixture-scale

contract now, while reserving large-instance exact performance for future solver transcripts and benchmark evidence.

References

BISECT project. U.17 branch-and-price redistricting paper plan, 2026. Local planning document and implementation roadmap.